

A Game-Native Model of the Extraspecial Extension

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Abstract

The flagship paper’s extraspecial dictionary identifies quadratic refinements Q of alternating forms B on $V = \mathbb{F}_2^n$ with central extensions $1 \rightarrow \mathbb{Z}/2 \rightarrow E_Q \rightarrow V \rightarrow 1$: squares in E_Q read the diagonal, commutators read the polar form. Its equivariance screen shows that once the extension is available as an object, the realizing rule layer is canonical – the squaring rule is the extraspecial squaring map read as a one-move relation – and its no-go theorems show the screen cannot be met by commutative value-worlds, by symmetry classes beyond the orthogonal group, or by E -translation dynamics. The flagship’s fourth residual open question is therefore structural: exhibit a *game-native* model of E_{Q_a} – positions built from game values, multiplication from game constructions – producing the Gold diagonal $q_i = Q_a(e_i)$ as squares, without an evaluation circuit for Q_a . A positive answer upgrades the flagship’s covariant realization to a structural one. This note fixes the demands such a model must meet, collects the proved fence, and records candidate directions: the mover-bit asymmetry as the central involution, noncommutative game compounds (ordinal sum and sequential join) as multiplication carriers, and the Brown-selector precedent that partizan outcome classes can host central data.

1 The dictionary and the problem

Throughout, $V = \mathbb{F}_2^n$, B alternating, Q a quadratic refinement of B , and $Q_a(x) = \text{Tr}(x x^{2^a})$ the Gold trace family on nimber fields that motivates everything [1].

Theorem 1 (extraspecial dictionary). **[STANDARD MATH]** *(proved self-contained in [1]; classical: [4, 2]) Central extensions $1 \rightarrow Z \rightarrow E \rightarrow V \rightarrow 1$ with $Z = \{1, z\} \cong \mathbb{Z}/2$ correspond bijectively to quadratic maps $Q: V \rightarrow \mathbb{F}_2$: lifted squares are central and well defined, giving $\tilde{x}^2 = z^{Q(x)}$, and lifted commutators give $[\tilde{x}, \tilde{y}] = z^{B(x,y)}$ with B the polar form of Q . E_Q is abelian iff $B = 0$, and extraspecial iff B is nondegenerate.*

Two structural consequences fix what any model must contain.

Lemma 2 (demands). **[STANDARD MATH]** *Any model (N, Z, π) of E_Q – a monoid N with central two-element subgroup Z and surjection $\pi: N \rightarrow V$ with fibers the Z -cosets – satisfies:*

- (i) if $B \neq 0$, N is noncommutative;
- (ii) if $Q(v) = 1$ for some v , any lift $\tilde{v}$ has order four, so N contains 4-torsion.

Proof. (i) is the dictionary: commutators read B . (ii): $\tilde{v}^2 = z \neq 1$ and $\tilde{v}^4 = z^2 = 1$. □

Question 3 (game-native extraspecial model). **[OPEN]** (the flagship’s stated open question) Does any game-native construction realize E_{Q_a} – positions built from game values, multiplication from game constructions – equivalently, produce the diagonal data $q_i = Q_a(e_i)$ as squares in a noncommutative game-built structure, without an evaluation circuit for Q_a ?

The payoff is canonicity: over the abstract group the flagship’s equivariance screen is already satisfiable by the squaring rule, which consults only the multiplication of E . A game-native E_{Q_a} would therefore upgrade the weighted-source realization, which is covariant (public preprocessing transported along the orthogonal group), to a structural one, where the arena *is* the extension.

2 The proved fence

Theorem 4 (commutative squaring collapse). **[PROVED]** [1, Lem. abelian, Thm. B] *In any commutative monoid, squaring is an endomorphism, so any model as in Lemma 2 inside a commutative monoid has linear squaring form: $B \equiv 0$. No commutative game-value monoid – Sprague–Grundy values under disjunctive sum, misère quotients, turn-parity counters, coin-turning configuration sums – hosts E_Q once $B \neq 0$.*

In concrete game terms the two demands of Lemma 2 fail separately for the default operations: disjunctive sum of impartial games is commutative *and* elementary abelian ($G + G$ is a P -position for every impartial G), so it lacks both the noncommutativity and the 4-torsion. The model needs a genuinely different multiplication.

Theorem 5 (the symmetry dial has no middle). **[PROVED]** [1, Thm. E] *By Dye–Dieudonné maximality of $O(Q)$ in $\text{Sp}(B)$ and the Witt orbital classification, a rule invariant under any group strictly larger than $O(Q)$ has dead P -set, while a fully $O(Q)$ - or $\text{Aut}(E)$ -equivariant rule factors extensionally through pointwise Q - and B -evaluation. E -naturality must be covariance, not invariance.*

Theorem 6 (E -translation no-go). **[PROVED]** [1, Thm. H] *For polar rank ≥ 4 , no left- E -equivariant rule on E_Q with moves $g \rightarrow gn$ for a fixed nonempty move set $N \subseteq E_Q$ has P -set the involution locus $\pi^{-1}(\{Q = 0\})$.*

So the model cannot be reached by making play *more* symmetric: the extension must be built, not imposed, and once built its group structure must drive the dynamics covariantly. The search space is exactly the noncommutative resources games natively possess.

Remark 7 (adjacent negative: the game-exterior collapse). **[PROVED]** [1, game-exterior appendix] For ambient-coherent coefficient-faithful Clifford data on the additive group of short partizan games, two-divisibility forces all quadratic and polar values into $\bigcap_k 4^k R$, hence zero over $\mathbb{Z}$, in characteristic 2, and over $\mathbb{Z}/4$. This kills the *additive, algebra-valued* route: one cannot source the extension’s quadratic data from a Clifford structure carried by the full game group. The extension target is group-theoretic rather than algebra-theoretic, so the collapse does not decide Question 3; it does warn that ambient coherence over the whole game group is fatal, and that a model should be sought over bounded-birthday or otherwise restricted position sets.

3 Resources and candidate directions

Remark 8 (the available asymmetry). The flagship identifies the one asymmetry that normal, misère, and partizan play all possess and that the symmetric polar form discards: the first-/second-player asymmetry of the move relation [1, equivariance-screen remark]. The central involution z

is the natural tenant: a tempo token whose consumption commutes with everything (centrality) but whose *presence* records one bit. Under this reading, lifted noncommutativity $[\tilde{x}, \tilde{y}] = z^{B(x,y)}$ says that interchanging the order of two position-building operations costs a tempo exactly when their polar pairing is on – a move-order holonomy. The FIFO charge mechanism of the flagship realization is precisely such a holonomy read *covariantly*; the model question asks for it *structurally*. [SPECULATION]

Remark 9 (noncommutative game compounds). [SPECULATION] Theorem 4 bars commutative operations, not game constructions as such. Games natively carry noncommutative compounds: ordinal sum $G : H$ (moves in H vanish when G is moved), sequential join, and side-conditional compounds. None is a group operation on values, so a model would quotient a compound-generated position monoid by an indistinguishability relation tuned so that the quotient is a finite 2-group with the right center – the analogue, one level up, of misère quotient construction, which Theorem 4 shows must *not* collapse the compound to a commutative value world. The first concrete probe: compute indistinguishability quotients of small ordinal-sum position algebras over the anisotropic plane ($n = 2$, $q_1 = q_2 = b_{12} = 1$, $E_Q = Q_8$) and check which quotients contain Q_8 ; the quaternion target makes the search finite and sharp.

Remark 10 (the Brown-selector precedent). The flagship’s Brown-selector theorem reads the $\mathbb{Z}/4$ -valued refinement off four partizan outcome classes $\mathcal{N}, \mathcal{R}, \mathcal{P}, \mathcal{L}$ with no external product or terminal evaluator [1, Brown-selector theorem]. This is evidence that partizan outcome observation has the capacity to host exactly the kind of central, order-four data the extension carries (Lemma 2(ii)): the selector’s outcome lattice is a $\mathbb{Z}/4$ readout obtained from game structure alone. What the selector does not provide is multiplication – it reads residues, it does not compose them. A model would need the composition law; the selector suggests where its values can live. [SPECULATION]

Remark 11 (implemented infrastructure). The repository implements the finite Heisenberg/Pauli representation of the char-2 extraspecial surface over its $\mathbb{F}_2$ bitmask boundary (`src/forms/char2/extraspecial.rs`): E_Q as explicit matrices, projective symplectic-transvection intertwiners, and the squaring/ commutator readouts. This is experiment infrastructure – a concrete target to compare candidate game quotients against – not a game-native model.

4 Target

Exhibit a set of positions generated by game constructions, a game-built composition law on them, a central two-element subgroup, and a surjection to the coordinate space, such that the squaring form is Q_a – with the diagonal produced by the structure, not queried – or prove a further no-go narrowing “game-native” (for instance: no finitely presented quotient of an ordinal-sum position monoid on game-value generators contains Q_8 compatibly with its Grundy projection). By Theorem 4 the source cannot be any commutative game-value monoid; by Lemma 2 it must carry 4-torsion; by Theorems 5 and 6 neither extra symmetry nor translation dynamics can substitute for the construction.

This note is the authoritative statement of the problem and will be folded into the flagship paper when resolved.

References

- [1] a9lim, *Quadratic refinements in normal play: realization and the observation boundary*, companion flagship paper, `writeups/goldarf.tex` in the ogdoad repository, 2026.

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- [3] T. E. Plambeck, A. N. Siegel, *Misere quotients for impartial games*, J. Combin. Theory Ser. A **115** (2008), 593–622.
- [4] D. Quillen, *The mod 2 cohomology rings of extra-special 2-groups and the spectra of quadratic forms*, Math. Ann. **194** (1971), 197–212.